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Title of paper : The Deep Learning Method Applied to The Detection and Mapping of Stone Deterioration in Open-Air Sanctuaries of The Hittite Period in Anatolia
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1. **What is the most important take home messages (conclusion) in the paper?**

A. Deep learning, using the Mask R-CNN algorithm, can automatically detect and map stone deterioration in archaeological heritage sites with fast and accurate detection.
B. The developed models, trained on images of Hattusa Yazılıkaya monuments, achieved high average precision values, ranging from 89.624% to 100% for different deterioration classes.
C. The models can be transformed into a practice that can be continuously monitored with mobile or computer applications which can be used effectively in all stone heritage sites and in which change can be followed quickly.

2. **What is general motivation for the research?**

A. The detection of deterioration in archaeological heritage sites is a very time-consuming task that requires expertise.
B. Vision-based approaches can cause errors, considering the complex types of deterioration that develop in different scales and forms in monuments.
C. There are deficiencies in the literature for determining the types of deterioration in archeological sites via computer vision.

3. **What is the research gap from previous work (technical problem that the authors found in previous literatures)?**

A. In previous studies, applications in which a limited amount of deterioration (spalling, efflorescence, and damage) was detected in building materials were made.
B. There are no studies on reliefs where detection is more difficult.
C. There are deficiencies in the literature for determining the types of deterioration in archeological sites via computer vision.

4. **What are solutions proposed by other researchers to solve the general research problem? What are disadvantages of these previous works?**

A. **Remote sensing, relief extraction, automatic drawing, and photogrammetry:** These methods were first used, but they have inadequacies in identifying deterioration.
B. **Terrestrial laser scanning and unmanned aerial vehicles with computer software:** Used for digital reconstruction and surface erosion examination but have inadequacies in identifying deterioration.
C. **Faster R-CNN algorithm:** Verschoof-van der Vaart and Lambers 1 identified prehistoric barrows and Celtic fields using the Faster R-CNN algorithm. Since object detection is carried out with the bounding box in this method, quite complex results can be obtained in cases with a high number of objects.

5. **In this paper, what is the proposed solution from the authors to solve the research gap and the general problem? According to the authors, why their proposed solution is important? Is there any improvement to previous works? How accurate the improvement?**

A. The authors propose using the Mask R-CNN algorithm to automatically detect and map deteriorations and restoration interventions.
B. Mask R-CNN has recently come to the fore with its feature of recognizing small and large-sized objects
C. The average precision values with high accuracy rates ranging from 89.624% to 100% were obtained for the deterioration classes.

6. **How did the authors validate their proposed solution? Is the solution supported by valid data or experimental results? Is there any discrepancy between what authors claimed and what they wrote in the paper?**

A. A total of 2460 images of Yazılıkaya monuments in the Hattusa archaeological site were gathered.
B. The developed algorithms were tested on 4 different rock reliefs in Yazılıkaya, which were not used in the training phase.
C. An image of the Eflatunpınar water monument, which is on the UNESCO tentative list, was used to test the model's universality.

7. **Try to analyze these parts: research gap, proposed solution, and how to evaluate the proposed solution. Do you find any problem from these three important parts? For example, in simulation-based research, do you think that the proposed solution can be implemented in real-time situation? Do you think that the proposed solution is robust and feasible for different setting/environment? How about the computational time?**

A. **Research Gap:**
   - Previous studies primarily focused on detecting limited types of deterioration (spalling, efflorescence, and damage) in building materials.
   - There is a lack of research on reliefs, where detection is more challenging.
   - The existing literature is deficient in determining the types of deterioration in archeological sites using computer vision techniques.
B. **Proposed Solution:** Automatic deterioration detection and mapping using the Mask R-CNN method in reliefs.
C. **Evaluation:** Training and testing the algorithm on Yazılıkaya reliefs and the Eflatunpınar water monument.

Potential problems:

A. The orthophotographs of the monuments were obtained using the PhotoScan program of the sequential images of the reliefs. To accelerate the training and test stages of the proposed model, a computer with a Graphics Processing Unit (GPU) processor and NVidia GeForce GTX 1060 GPU graphic card was used.
B. The types of deterioration developing in stones used in historic monuments as a result of atmospheric, anthropogenic, and biological processes have been classified by many researchers.
C. The image of the Eflatunpinar water monument built in a different climatic environment and lithology was used

8. **What is scientific contribution in this paper? Did the authors solve old problem with new approach? Did the authors solve new problem with old approach? Did the authors solve new problem with new approach?**

A. The algorithms developed within the scope of the study are the first model with the feature of Mask R-CNN-based deterioration identification and mapping in stone heritage sites.
B. The model was tested in monuments in two different regions belonging to different environmental conditions, stone type and construction technique, and its universal usability was proven.
C. It is obvious that it can easily be transformed into a practice that can be continuously monitored with mobile or computer applications which can be used effectively in all stone heritage sites and in which change can be followed quickly

9. Try to read the discussion and conclusion. Do you find any suggestion for improvement in the discussion? What are disadvantages of the proposed solution?

A. The time of experts in the field can be optimized by the autonomous UAVs to be integrated into this method, which is extremely fast in deterioration detection and mapping, in future studies.
B. A few minor errors were found in the model. The first one of these errors is that the crack type not observed in the structure was detected in the model.
C. The other problem of detection is that only one of the missing part types found in three different parts of the monument could be detected.

10. Do you find any part of this paper that you don't understand clearly? (e.g.: mathematical equation, algorithm of the proposed solution, validation steps, etc.). Try to find as many as possible parts that you don't understand. This list will guide you to study basic knowledge needed to better understand the paper.

A. **The Mask R-CNN Algorithm:** Understanding the specifics of how the Mask R-CNN algorithm functions. This includes delving into the underlying Convolutional Neural Networks (CNNs), the Region Proposal Network (RPN), and how these components work together to achieve object detection and instance segmentation.
B. **Network Head:** The network head is responsible for classifying objects, refining bounding box predictions, and generating segmentation masks. Further understanding of how classes, bounding boxes, and fixed-size representations are handled within the network head would be beneficial.
C. **Implementation Details:** Grasping the implementation details, such as the specific software libraries used (TensorFlow framework2), the hardware used (GPU processor and NVidia GeForce GTX 1060 GPU graphic card2), and the training parameters is important for replicating or adapting the research.

RESEARCH COMPARATIVE TABLE BETWEEN 10 JOURNALS AND PROPOSED RESEARCH

AZAM AZRI AHMAD / 24/552221/PTK/16448

Title	Author	Year	Main Focus	Scope Of Work	Key Findings	Methodologies	Key Practices	Algorithm
Classification of fragments: recognition of artistic style	Lucia Cascone et al.	2022	Classification of artistic fragments using ML	Analysis of different ML algorithms on fragmented fresco datasets	Random Forest outperforms SVM and KNN; highest accuracy of 34.42% with 20 fragments	Experimental analysis with datasets and ML algorithms	Use of shape, color, and texture features for classification	Random Forest, SVM, KNN
Digital Twins for Protecting Cultural Heritage Against Climate Change	Mylonas, G., Kalogeras, A., Pavlidis, G., Lalos, A., & García-López, A.	2023	Use of digital twins and cyber-physical systems for cultural heritage protection	Monitoring and protecting cultural heritage sites from climate change impacts	Digital twins enhance protection against climate-related threats	Integration of real-time data, AI, and IoT technologies	Degradation detection and maintenance optimization for cultural heritage sites	Digital twin algorithm
Improving Cultural Heritage conservation: LSTM neural networks to effectively processing end-user's maintenance requests	Marco D'Orazio, Gabriele Bernardini, Elisa Di Giuseppe	2023	Preventive conservation of cultural heritage buildings	Involves analyzing data collected from a computer maintenance management software (CMMS) for a cultural heritage building in Italy, specifically the Villarey Barrack in Ancona	Prediction accuracy of 96.6% in categorizing maintenance requests	LSTM neural networks, analysis of maintenance requests from CMMS	Adapting to emerging issues, reducing invasive interventions	LSTM
Evaluating Activation Functions in GAN	Maitin, A. M., Nogales, A.,	2024	Performance evaluation of	Restoration of Greek temples	ReLU outperforms others; Mish and	Use of GANs for virtual	Direct training without segmented	Generative Adversarial

Models for Virtual Inpainting: A Path to Architectural Heritage Restoration	Delgado-Martos, E., Intra Sidola, G., Pesqueira-Calvo, C., Furnieles, G., & García-Tejedor, Á. J.		activation functions in GANs for virtual inpainting		Leaky ReLU perform poorly; Swish close to ReLU in subjective evaluations	inpainting; objective and subjective evaluations	images to reduce computational demands	Network (GAN)
Artificial intelligence-assisted visual inspection for cultural heritage: State-of-the-art review	Mayank Mishra, Paulo B. Lourenço	2024	AI-assisted inspections for damage detection in cultural heritage structures	Cultural heritage structures	AI technologies can automate inspections, increase accuracy, and integrate with VR for training	Deep learning techniques, computer vision, integration with non-destructive testing	Use of AI for automated inspections, digitization of inspection stages	YOLO, CNNs, integration with VR and NDT tools
Application of computer vision technology in surface damage detection and analysis of shedthin tiles in China: a case study of the classical gardens of Suzhou	Yan et al.	2024	Enhancing efficiency and accuracy in detecting damage types in shedthin tiles for cultural relic protection	Damage assessment and restoration of shedthin tiles in historical gardens in Suzhou, China	1. Model with minimum validation loss (90 epochs) achieved better detection accuracy. 2. Struggled with distinguishing between surface scaling and water stains. 3. Demonstrated robustness in real-world scenarios.	Training YOLOv4 model with 500 images, using a freezing stage (10 epochs) and an unfreezing stage (750 epochs)	Data preprocessing, annotation, data augmentation, real-time damage detection	YOLOv4
New technologies for	María Belén	2023	Impact of AI and	Analysis of 1,646	Significant increase	Bibliometric	Identification of	Not

the conservation and preservation of cultural heritage through a bibliometric analysis	Prados-Peña, George Pavlidis, Ana García-López Alonso Cano		ML on heritage conservation and preservation	articles from Scopus database	in AI/ML literature; limited publications specifically on cultural heritage; need for further research	analysis	trends, international cooperation, keyword co-occurrence analysis	specified
Integrating Emerging Technologies with Digital Twins for Heritage Building Conservation: An Interdisciplinary Approach with Expert Insights and Bibliometric Analysis	Silvia Mazzetto	2024	Integration of emerging technologies in heritage conservation	Conservation of heritage buildings using DTs, BIM, ML, IoT	Shift towards preventive conservation; need for standardized protocols; potential of AR and VR	Literature review, expert interviews, bibliometric analysis	Multidisciplinary approach, real-time monitoring, digital representations	Not specified
Automated Evaluation of Degradation in Stone Heritage Structures Utilizing Deep Vision in Synthetic and Real-Time Environments	T.J. Saravanan and B. Bhaskar	2024	Automated defect detection in stone heritage structures	Detection and classification of defects such as vegetation growth and cracks using deep learning	ResNet101 achieves higher accuracy (94%) compared to ResNet50 (93%)	Mask R-CNN with ResNet50 and ResNet101 backbones, using Adam's optimizer	Use of synthetic images generated with Blender 3D to enhance dataset diversity	Mask R-CNN
Intelligent Detection of Deterioration in Cultural Stone Heritage	M. Ergün Hatır, İsmail İnce, Mustafa Korkanç	2021	Detection and mapping of deterioration in cultural heritage structures	Involves the application of a deep learning-based Mask R-CNN model to detect and map deterioration in the Gümüşler	High accuracy in identifying deterioration types with mAP of 98.186%	Deep learning-based Mask R-CNN model trained on 26,000 images	Use of drone-captured images for training; real-time monitoring potential	Mask R-CNN

				Monastery, a complex historical building in Turkey.				